More AI Reviewers Help — *Only If Their Mistakes Differ*

Companion to: "Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study"

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01 MOTIVATION

LLMs increasingly automate code review, and frameworks split the work across specialized agents — managers, specialists, voters — and price accordingly.

Does the *structure* of that collaboration improve review quality, or just cost more? We hold the model, PR snapshot, prompts, and pipeline fixed and vary only structure. Pre-registered on OSF; prompts frozen before any confirmatory data.

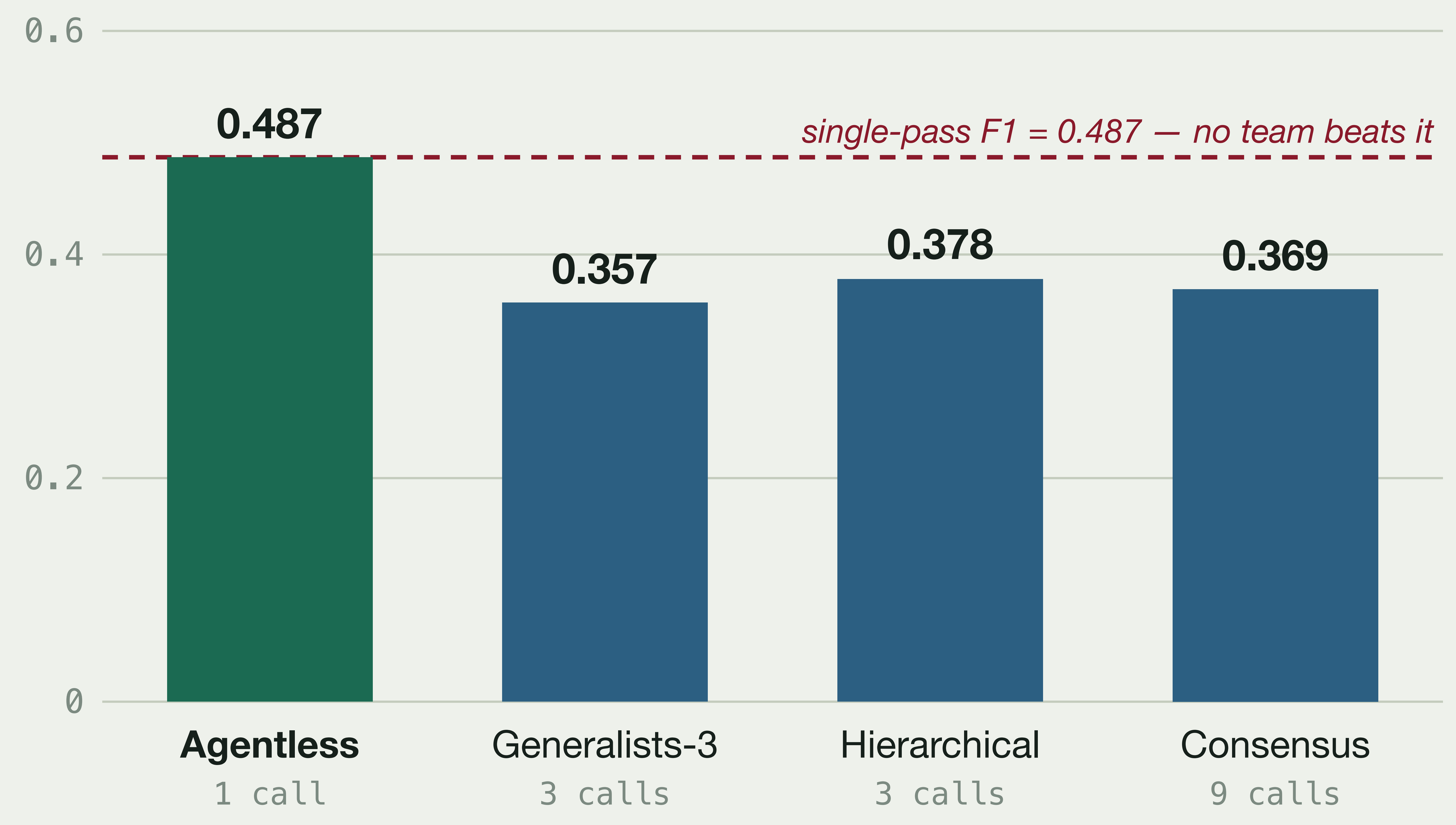
02 METHOD — A FOUR-RUNG LADDER, EACH RUNG ADDS ONE CAPABILITY

Agentless `single pass · 1 call` → **Generalists-3** `sample ×3, merge · 3 calls` → **Hierarchical** `coordinator + specialists · 3 calls` → **Consensus** `debate + vote · 9 calls`

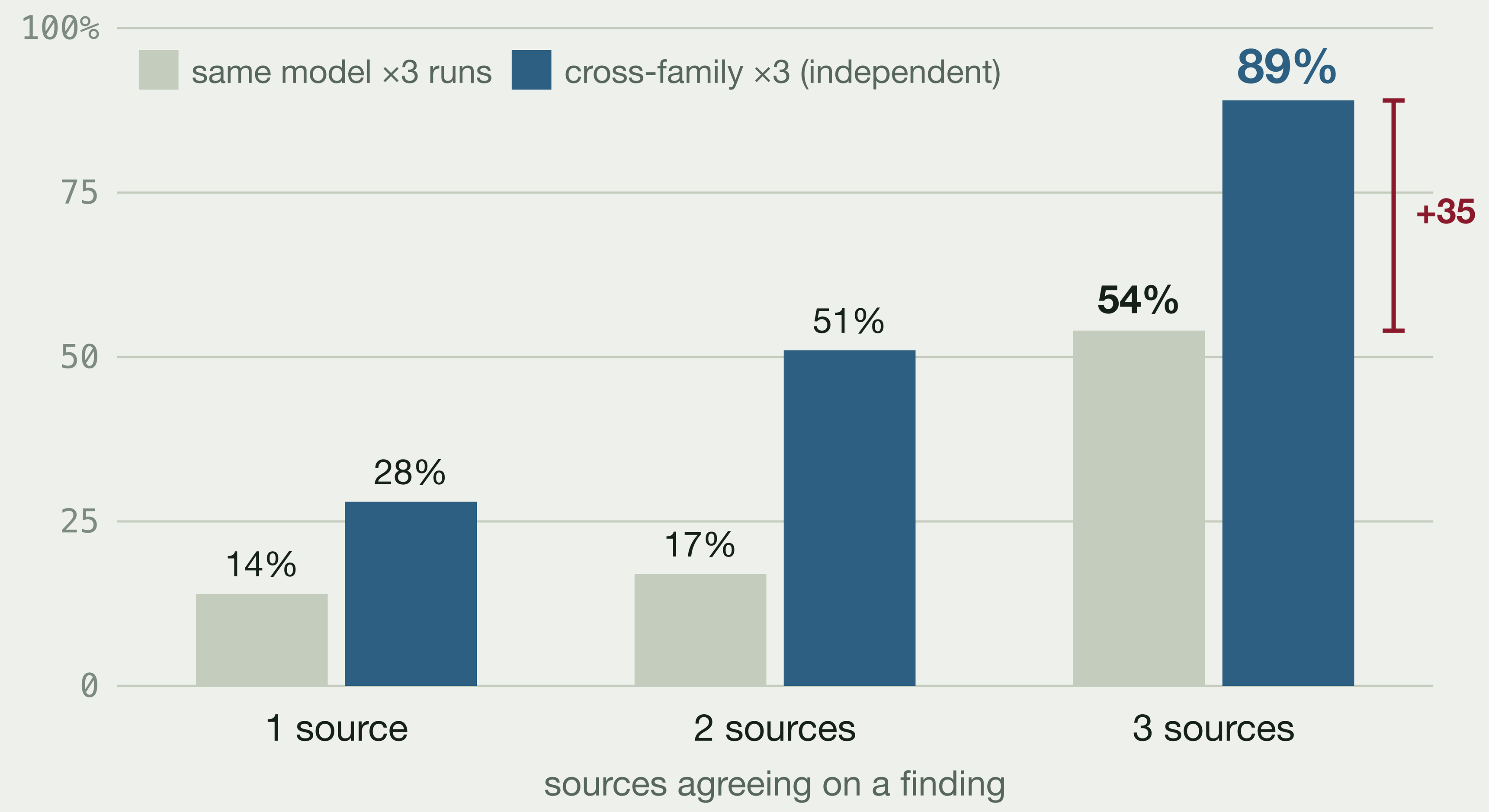
Adjacent contrasts isolate **sampling** vs **specialization** vs **communication**. Model held constant: Claude Haiku 4.5. Evaluated on three datasets:

- **E1 · Qodo** `99 PRs · 1,188 reviews · injected ground truth`
- **E2 · SWE-PRBench** `50 PRs · 600 reviews · human-reviewer agreement`
- **E3 · Industrial** `30 PRs · 627 reviews · no oracle`

03 TWO RESULTS: MORE AGENTS DON'T HELP — DECORRELATION DOES



The null (myth-buster). No multi-agent arm beats single-pass F1. Extra agents raise recall (0.62–0.63 vs 0.50) but halve precision — buying recall at **3–9× the calls** and **5× the cost** per confirmed finding.



The signal (decorrelation). Three *independent* model families match injected ground truth **89%** of the time; one model repeated three times saturates at **54%**. Independence — not more compute — is what makes agreement predict truth.

04 KEY FINDINGS (PRE-REGISTERED, CONFIRMATORY)

- **Specialization is null:** Hierarchical ties Generalists-3 on recall (0.624 vs 0.626, $p=0.73$).
- **No arm beats single-pass F1** (0.487 vs 0.357–0.378; Holm $p<0.001$).
- **Cost rises monotonically** along the ladder — efficiency, not quality.
- **Communication unsupported:** Consensus does not beat Hierarchical (Holm $p=0.12$) at 5× the cost.
- **Grounding bounds recall at a ≈0.83 ceiling** — linter-recoverable conventions + an execution-reachable functional core.
- **Self-consistency fails; cross-family agreement lifts precision.**

06 WHY IT MATTERS

- Don't deploy multi-agent review expecting a quality lift — you buy recall, not F1, at 3–9× the cost.
- Decorrelate verifiers: independent model families, not one model repeating itself.
- Measure benchmark completeness — a 0.83 recall ceiling means much "imprecision" is the benchmark's.
- Treat LLM-judged evaluation on real code as unreliable ($\kappa = 0.03$).

07 NEXT STEPS · OPEN SCIENCE

- Tool-grounded verifiers — execution + static analysis as a correctness oracle.
- All runs, ablations, and artifacts released for replay.

05 WHERE RESEARCH MEETS THE CLIENT: E3 IS THE INDIGENOMICS RAP PLATFORM

Our E3 industrial dataset *is* the live RAP data platform we built for the **Indigenomics Institute** — its **30 real merged pull requests, 627 reviews**. On this unlabeled industrial code the verification relationship reaches its *ecological boundary*: LLM correctness judges themselves disagree (**$\kappa = 0.03$**), so the decorrelation signal that holds on labeled benchmarks cannot be validated without an oracle. The research and the client's product are one system.

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Pre-registered on OSF · prompts frozen before confirmatory data

github.com/SharonHuang77/rap-review-research